Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 5: Turning an Implication Around

Converse, Contrapositive, and the Language of "If"

Chartrand & Zhang, Section 1.3, pp. 28–32

Name: _____ Date: _____

Where we are

Monday you saw that $P(x) : x = 3$ and $Q(x) : x^2 = 9$ give an implication that is true for every real number — but reversing it produces one that fails at $x = -3$. Today we name that reversal, find the *other* rearrangement that does not change anything, and learn the six English phrasings that all mean the same implication.

By the end of today you should be able to

1. Write the converse, contrapositive, and inverse of an implication.
2. Say which of the four are logically equivalent to each other.
3. Translate "only if," "sufficient," and "necessary" into "if…then."

1. Six ways to say the same thing

Figure 1.10 — expressing $P \Rightarrow Q$ in words

If P, then Q.
Q if P.
P implies Q.
P **only if** Q.
P is **sufficient** for Q.
Q is **necessary** for P.

"Only if" is the one that catches people

A sign-up sheet reads: *"Sign this sheet only if you plan to take discrete mathematics next semester."* What it means: if you sign the sheet, then you plan to take the course.
Note the direction. "P only if Q" is $P \Rightarrow Q$, not $Q \Rightarrow P$. The phrase names the *necessary condition* second, but the arrow still points from P.

Necessary vs. sufficient. For $P \Rightarrow Q$:

- P is **sufficient** for Q — having P is *enough* to guarantee Q.
- Q is **necessary** for P — without Q you cannot have P.

Rewrite each as "if… then."

Given	If… then
Passing the final is sufficient to pass the course.	If you pass the final, then you pass the course.
A valid token is necessary to access the file.	If you access the file, then you have a valid token.
The program compiles only if the syntax is correct.	If the program compiles, then the syntax is correct.

2. Three rearrangements

Start with $P \Rightarrow Q$. There are three natural things to do to it.

Definitions 1.41 and 1.44

Name	Form	Built by
Implication	$P \Rightarrow Q$	—
Converse	$Q \Rightarrow P$	swapping
Inverse	$(\sim P) \Rightarrow (\sim Q)$	negating both
Contrapositive	$(\sim Q) \Rightarrow (\sim P)$	swapping *and* negating

Build the table and find out which agree.

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$(\sim P) \Rightarrow (\sim Q)$	$(\sim Q) \Rightarrow (\sim P)$
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	T	T	F	F	T
F	F	T	T	T	T

Which column matches the implication? the contrapositive

Which two columns match each other? converse and inverse

Theorem 1.46

$$P \Rightarrow Q \;\equiv\; (\sim Q) \Rightarrow (\sim P)$$

An implication and its contrapositive always have the same truth value. An implication and its converse need not.

Why the contrapositive earns its own theorem

When a direct proof of $P \Rightarrow Q$ is awkward, you may instead prove $(\sim Q) \Rightarrow (\sim P)$ — and you will have proved the original, because they are the same statement. This is **proof by contrapositive**, and it is one of the main techniques of Chapter 3.
Example: proving "if n^2 is even then n is even" directly is unpleasant. Proving "if n is odd then n^2 is odd" is three lines.

3. Practise the four forms

Take: *If a number is divisible by* 4*, then it is even.*

Form	Statement	True?
Implication	If a number is divisible by 4, then it is even.	Yes
Converse	If a number is even, then it is divisible by 4.	No (6)
Inverse	If a number is not divisible by 4, then it is not even.	No (6)
Contrapositive	If a number is not even, then it is not divisible by 4.	Yes

Notice the pattern of answers: the implication and contrapositive agree, the converse and inverse agree, and **one counterexample refutes both** of the middle two. Here that counterexample is $n = 6$.

Confusing an implication with its converse

This is the most consequential error in the whole chapter, and it is not confined to mathematics. "If you have the disease, the test is positive" and "if the test is positive, you have the disease" are different claims, and a test can satisfy the first while badly failing the second. Assuming the converse follows is a genuine reasoning error with real consequences.

Exit ticket

1. Write the contrapositive of "If the list is sorted, then the list has no inversions."

2. "You may enrol only if you have taken CS 204." Rewrite as "if…then."

3. Which is logically equivalent to $P \Rightarrow Q$: its converse or its contrapositive?

Before Friday

Read Section 1.4, "Biconditionals," pp. 36–39.

Self-check (not collected). Section 1.3, Exercises 5, 7, 13, and 15. All odd-numbered — answers are in the back, so attempt them fully first.

Homework 2 is due Friday, August 28, 11:59 p.m.

One PDF to Canvas, with the completed collaboration and AI statement first.

Questions I still have